

EasyDriver v4.5

An easy to use bipolar stepper motor driver
Use 4 wire, 6 wire or 8 wire stepper motors
From about 150mA/phase to about 750mA/phase
Defaults to 5V for Vcc (logic supply), settable to 3.3V
Supply 8V to 30V DC power input on JP1
Do not connect or disconnect motor while EasyDriver is powered

DEFAULT OPTIONS

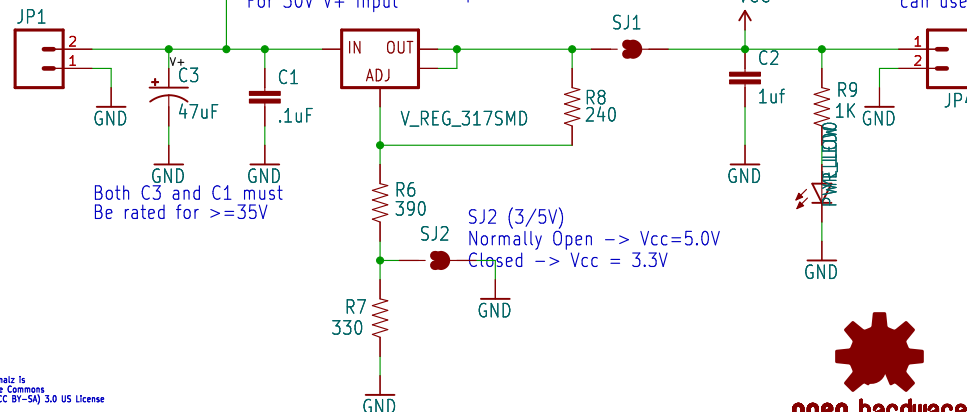
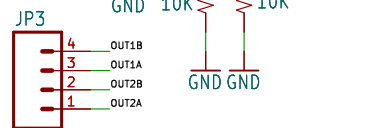
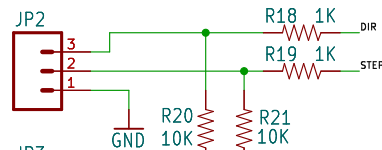
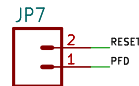
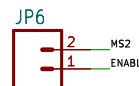
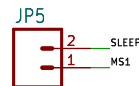
Short JP5, JP6, JP7 pins to GND or Vcc to override

SLEEP = Vcc (awake)
MS1 = Vcc (1/8 microstep)
MS2 = Vcc (1/8 microstep)
ENABLE = GND (enabled)
RESET = Vcc (not reset)
PFD = Vcc (slow decay mode)

DIR is level sensitive
A rising edge on STEP causes a step
Both take 0V to Vcc

Coil 1 of motor across
OUT1B and OUT1A
Coil 2 of motor across
OUT2B and OUT2A

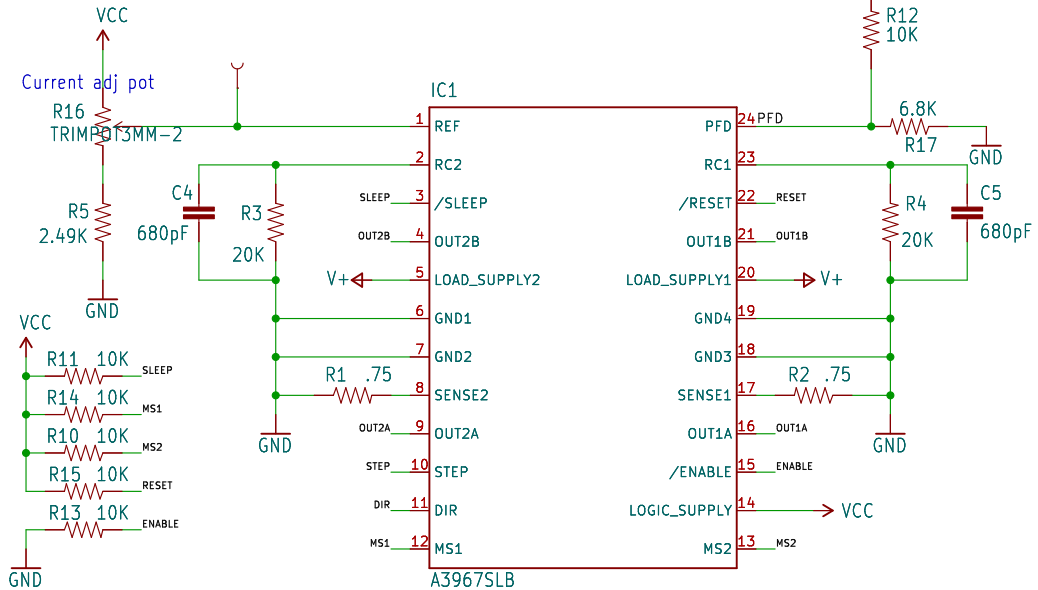
Power Input
8V to 30V (Vcc = 5V)
6.3V to 30V (Vcc = 3.3V)



TP1 = Vref input to driver
Monitor this test point with meter
as you adjust current adj pot
Valid range 1.0V to Vcc
At Vref of 5V max current will be 833mA
At Vref of 2V max current will be 333mA
At Vref of 1V max current will be 166mA
Minimum current gives smoothest microsteps
Maximum current gives highest torque
Max Coil Current(in Amps) = Vref(in Volts)/6
Set R16 to 2.0V at factory = 333mA/phase

www.schmalzhaus.com/EasyDriver

PFD intermediate voltage
Set for 'mixed-decay' mode.



Change List:
v4.3 12/09/09 BPS Added mounting holes
v4.4 10/24/20 BPS Fixed MIN/MAX silkscreen
All v4.4 now .02"
v4.4 1/2/12 BPS: C3 now 47uF
v4.5 2/25/14 BPS Added series resistors and pull downs on STEP/DIR
Updated pot footprint
Made bottom layer GND rather than VCC
Updated to latest SparkFun footprints
Switched from CC BY 3.0 to CC BY-SA 3.0 license
Fixed pot silkscreen direction
R5 now 2.5K Ohms to get down to 1V on Vref
R17 now 6.8K to set PFD to 'mixed-decay' mode

Released under the Creative Commons
Attribution Share-Alike 3.0 License
<https://creativecommons.org/licenses/by-sa/3.0/>

TITLE: \${PROJECTNAME}	SFE
Design by: Brian Schmalz	REV: 4.5
Produced by Spark Fun Electronics	
Date: \${CURRENT_DATE}	Sheet: \${#}/\${##}

Size: User Date: Rev: 1/1
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1

EasyDriver by Brian Schmalz is
licensed under a Creative Commons
Attribution Share Alike (CC BY-SA) 3.0 US License
Copyright 2010-2014

